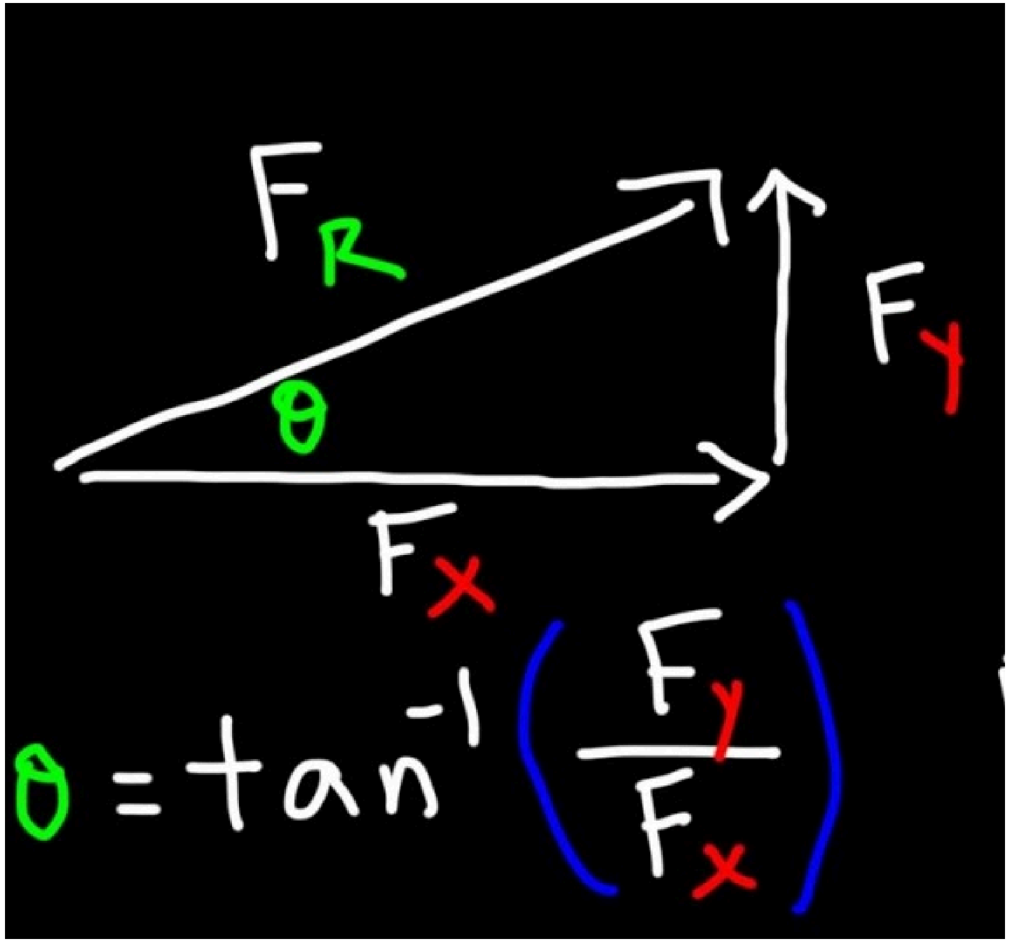
Day 04 - Mathematical Preliminaries

Questions? *Make sure to upvote questions*



RaiseMyHand

- Tool should be available for each class period.
- Answers to your questions are also written up after class
    - Day 02 - https://raisemyhand.msucerl.org/student?code=s97Hf7-WS9K7b7ooovdjKxXwih1b_Nnj
- Links to each day posted in MS Teams

Would appreciate your feedback and/or ideas for use cases

Announcements

- Homework 1 is due this Friday
- Homework 2 is posted now
- Help sessions start this week
    - DC Friday at 2-4pm (1248 BPS)
- Mihir (ULA) will host additional help hours soon

Seminars this week (Wednesday)

Goals for this week

Be able to answer the following questions.

- What are the essential physics models for single particles?
- How do we setup problems in classical mechanics?
- What mathematics do we need to get started?
- How do we solve the equations of motion?

Reminders from Day 03

- In a Newtonian world, we start from a vector description of motion
- Differential equations are mathematical models that describe the motion of particles
- We can use different methods to solve these differential equations

i-Clicker: https://join.iclicker.com/PRJO

Clicker Question 4-1

I feel confident in my abilities to use VS Code for my homework

1. Strongly Agree
2. Agree
3. We'll see
4. Disagree
5. Strongly disagree

Projectile Motion

$$\mathbf{a} = \langle a_x, a_y \rangle$$

$$x_f = x_i + v_{x,i}t + \frac{1}{2}a_x t^2$$

$$y_f = y_i + v_{y,i}t + \frac{1}{2}a_y t^2$$

Clicker Question 4-2

For this fountain, what is the best guess for the acceleration ($\mathbf{a} = ??$) experienced by a fluid particle?

Assume y is positive upward; x is positive to the right.

1. $a_x \neq 0, a_y = g$
2. $a_x = 0, a_y = g$
3. $a_x \neq 0, a_y = -g$
4. $a_x = 0, a_y = -g$
5. Something else

Clicker Question 4-3

I feel comfortable with a discrete formulation of Newton's Laws.

1. Yes, I got this.
2. I recall some ideas, but let's check in.
3. I'm not sure.
4. I really don't know what *discrete formulation* means here
5. ???

Clicker Question 4-4

The average velocity for a macroscopic time step $\Delta t = t_f - t_i$ is given by:

$$\mathbf{v}_{avg} = \frac{\Delta \mathbf{r}}{dt}$$

where $\Delta \mathbf{r} = \mathbf{r}_f - \mathbf{r}_i$. At what time do we estimate the average velocity occurs?

1. t_i
2. t_f
3. Sometime between t_f and t_i
4. $\dfrac{t_f - t_i}{2}$

Clicker Question 4-5

I feel comfortable with vectors, vector decomposition, and trigonometry in Cartesian coordinates.

1. Yes, I got this.
2. I recall some ideas, but let's check in.
3. I'm not sure.
4. I don't feel too confident with vectors.

Clicker Question 4-6

Consider the generic position vector $\vec{R}$ for a particle in 2D space. Which of the following describes the direction of the vector in plane polar coordinates (r, ϕ)?

1. $\hat{R}$
2. $\hat{r}$
3. $\hat{\phi}$
4. Some combination of $\hat{r}$ and $\hat{\phi}$
5. I'm not sure.

Group Discussion 4-1

We found the following expression for the equation of motion of a falling ball subject to air resistance:

$$m\ddot{y} = +mg - b\dot{y} - c\dot{y}^2$$

What are the units of the constants b and c?

Group Discussion 4-2

Consider the generic position vector $\vec{R}$ for a particle in 2D space. Find the velocity vector $\vec{V}$ for the particle in Cartesian coordinates (x, y).

What happens in plane polar coordinates?

$$\vec{R} = r\hat{r} + \phi\hat{\phi}$$

Note:

$$\hat{r} = \cos(\phi)\hat{x} + \sin(\phi)\hat{y}$$

$$\hat{\phi} = -\sin(\phi)\hat{x} + \cos(\phi)\hat{y}$$